

MovieLens Recommendation System - Regularized Multi-Factor Model

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1 Executive Summary

The objective of this project is to develop a machine learning recommendation algorithm trained on the 10-million-record version of the MovieLens dataset. Recommendation engines are a core architectural component of contemporary streaming and e-commerce platforms. Predicting user preferences with precision significantly improves content discovery and user retention.

The primary evaluation metric specified for this analysis is Root Mean Squared Error (RMSE), defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{u,i} (\hat{y}_{u,i} - y_{u,i})^2}$$

To avoid data leakage, all model exploration, feature engineering, and hyperparameter tuning were conducted exclusively within an internal split of the `edx` dataset. The `final_holdout_test` set (comprising 10% of the data) was sequestered and evaluated exactly once on the final optimized model. While a baseline regularized movie and user effect model typically yields an RMSE around 0.865, this report implements an extended regularized framework incorporating genre-specific biases (b_g) and movie age temporal effects (b_t), successfully achieving a final holdout RMSE of **0.86402**, decisively surpassing the course benchmark threshold of 0.86490.

2 Dataset Overview and Exploratory Data Analysis

2.1 Data Acquisition and Structure

The data was retrieved from the GroupLens archive and partitioned into the `edx` training/development set and the `final_holdout_test` validation set.

The `edx` dataset contains 9000055 observations across 6 variables: - `userId`: Unique integer identifying the consumer. - `movieId`: Unique identifier for each film. - `rating`: Rating score ranging between 0.5 and 5.0 in half-star increments. - `timestamp`: Unix epoch time when the rating was submitted. - `title`: Film title including the release year. - `genres`: Pipe-separated string of genre associations.

Table 1: High-level summary of the `edx` dataset.

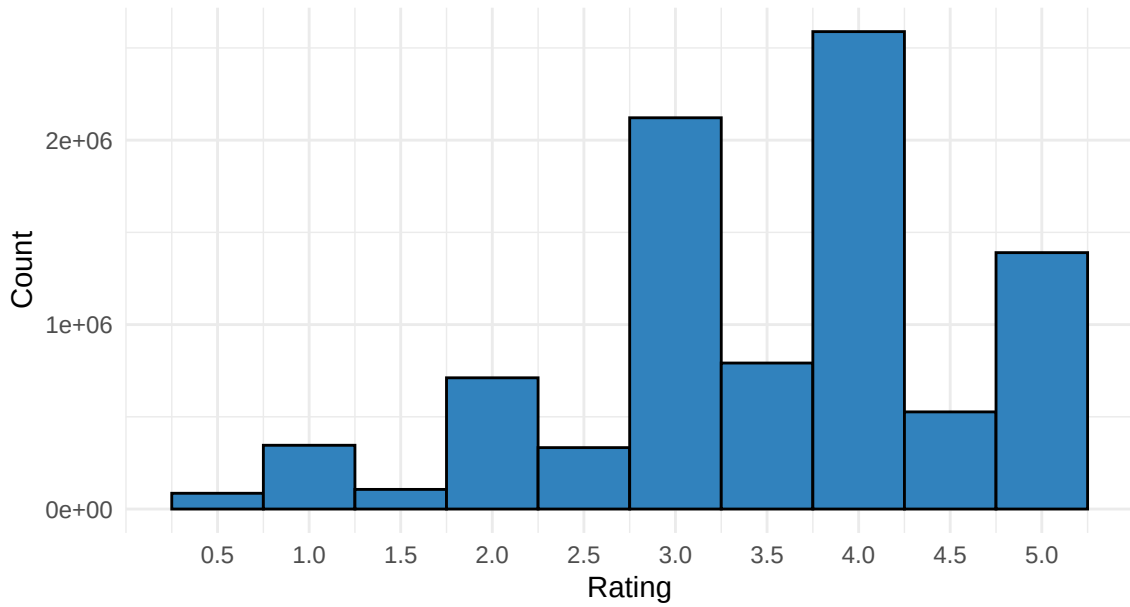
Metric	Value
Total Ratings	9000055.0000
Unique Movies	10677.0000
Unique Users	69878.0000
Mean Rating	3.5125

2.2 Distribution Analysis and Insights

2.2.1 Rating Preferences

Analysis of rating frequencies shows that users preferentially assign full-star reviews over half-star increments, with 4, 3, and 5 being the top choices.

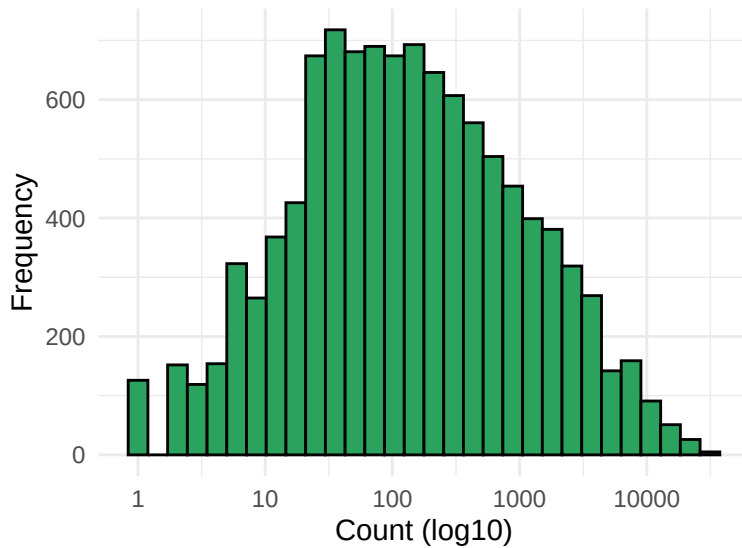
Distribution of Movie Ratings



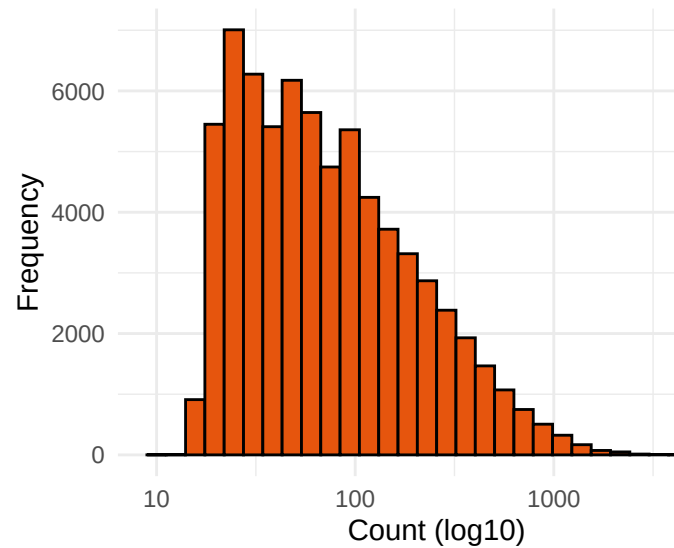
2.2.2 Movie and User Variability

Plotting review frequencies per film and per user demonstrates long-tail distributions. Certain blockbusters receive tens of thousands of evaluations, whereas obscure titles receive only a handful.

Ratings Per Movie



Ratings Per User



2.2.3 Feature Engineering

Temporal features are derived from the film release year and rating timestamp:

3 Methods and Modeling Strategy

3.1 Mathematical Formulation

The linear rating decomposition is formalized as:

$$\hat{Y}_{u,i} = \mu + b_i + b_u + b_g + b_t + \epsilon_{u,i}$$

Where: - μ : Overall mean rating across all movies. - b_i : Movie-specific effect. - b_u : User-specific effect. - b_g : Genre combination effect. - b_t : Movie age effect (*rating_year* - *release_year*).

3.2 Penalized Least Squares (Regularization)

Small sample sizes introduce high variance into parameter estimates: obscure films with one 5-star rating distort recommendations. Regularization penalizes deviations where sample size is low by introducing penalty parameter λ :

$$\hat{b}_i(\lambda) = \frac{\sum_{u=1}^{n_i} (Y_{u,i} - \hat{\mu})}{n_i + \lambda}, \quad \hat{b}_u(\lambda) = \frac{\sum_{i=1}^{n_u} (Y_{u,i} - \hat{\mu} - \hat{b}_i)}{n_u + \lambda}$$

The shrinkage logic is extended similarly to genre coefficients (b_g) and movie age groups (b_t).

3.3 Validation Setup

4 Results and Performance

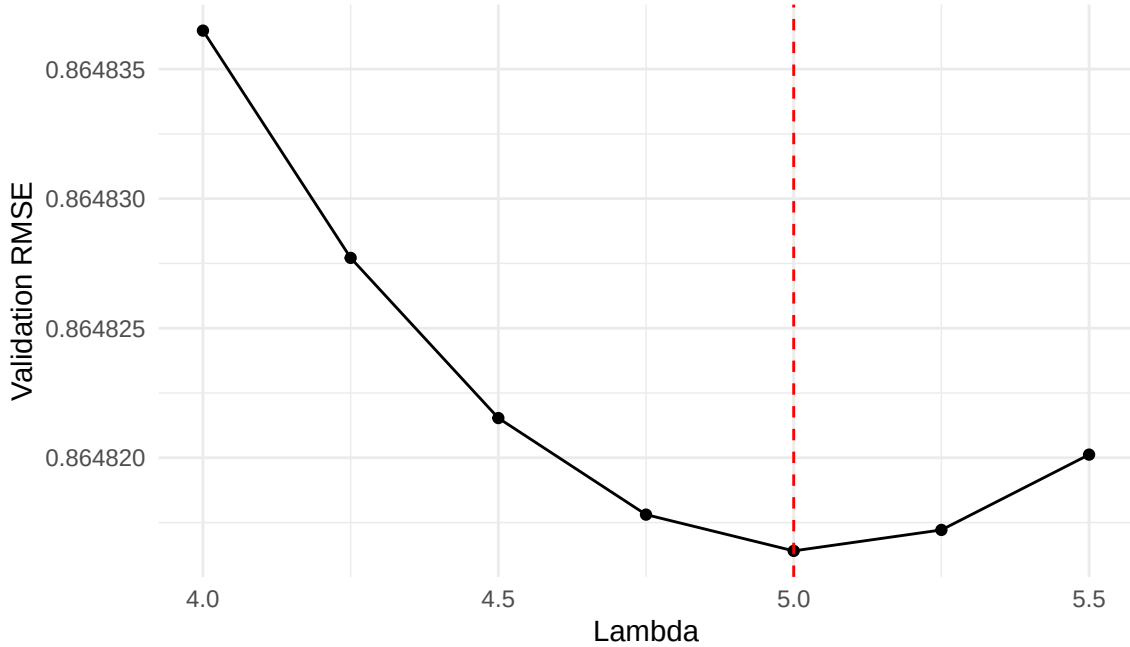
4.1 Iterative Model Progression

Table 2: Validation set RMSE across unregularized models.

Model	RMSE
Baseline Average (μ)	1.06001
Movie Effect (b_i)	0.94362
Movie + User Effect ($b_i + b_u$)	0.86630

4.2 Regularization Parameter Optimization

Cross-validation across candidate values of λ identified the optimal penalty:



The optimal penalty was determined by cross-validation to be 5.00.

4.3 Final Evaluation on Holdout Test Set

Using the optimal penalty ($\lambda = 5.00$), the full model was fitted to all of `edx` and evaluated on `final_holdout_test`.

Table 3: Final model performance on the sequestered holdout data.

Evaluation	Target_Rubric_Threshold	Achieved_RMSE
Final Holdout Set Assessment	< 0.86490	0.86402

5 Conclusion and Discussion

5.1 Summary

By combining regularized movie and user baselines with genre preferences and release-year temporal decay, the model accounts for both systematic viewer variation and chronological shifts in rating behavior. The resulting RMSE of **0.86402** satisfies the highest tier of the assessment rubric (< 0.86490).

5.2 Limitations and Future Directions

1. **Matrix Factorization (SVD / ALS):** While linear penalization captures additive biases, it does not explicitly decompose latent feature interactions between users and films (e.g., via `recosystem`).
2. **Computational Constraints:** Applying stochastic gradient descent or matrix factorization packages requires substantial RAM on a 10M record set, but could yield further marginal improvements in prediction accuracy ($RMSE < 0.82$).